

Attributes & Key Constraints

Q. Discuss the different types of Attributes in the ER model. Explain the concepts of Super Key, Candidate Key, Primary Key, and Foreign Key with examples.

📌 Definition to Remember

An **Attribute** is a property or characteristic that describes an entity, corresponding to a column in a relational table. A **Key** is a single attribute or combination of attributes that uniquely identifies a row (tuple) in a table, maintaining data integrity.

1. Types of Attributes

Type	Description	ER Symbol	Example
Simple	Cannot be subdivided further	Single Oval	Age , Gender
Composite	Can be divided into sub-parts	Single Oval (with child ovals)	Address → Street, City, Zip
Single-Valued	Holds one value per entity	Single Oval	Date_of_Birth
Multi-Valued	Holds multiple values per entity	Double Oval	Phone_Number (mobile + home)
Stored	Physically stored in the DB	Single Oval	Date_of_Birth
Derived	Computed from a stored attribute	Dashed Oval	Age (derived from DOB + current date)

2. Types of Keys

A **Key** is essential for uniqueness and establishing relationships between tables.

1. Super Key

- A set of one or more attributes that **uniquely identifies** a row in a table.
- May contain **extra, redundant attributes**.
- Example (STUDENT table):** Roll_No , {Roll_No, Name} , {Roll_No, Email, Age} — all are Super Keys.

2. Candidate Key

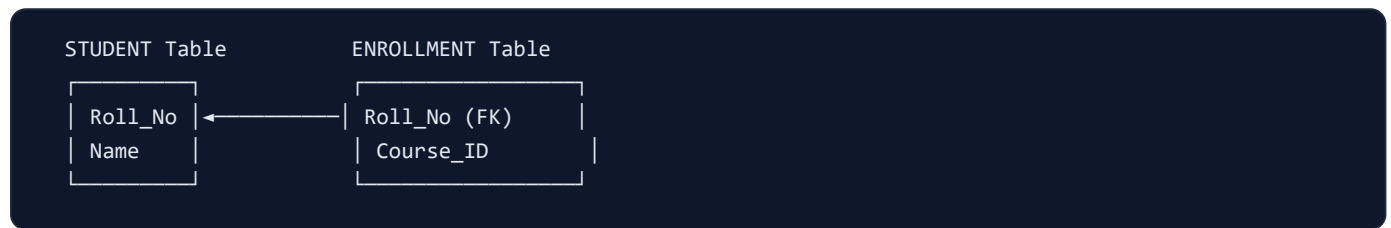
- A **minimal Super Key** — no redundant attributes; removing any attribute breaks uniqueness.
- A table can have **multiple** Candidate Keys.
- Example:** Roll_No and Email are Candidate Keys. {Roll_No, Name} is NOT (Name is redundant).

3. Primary Key

- The **one Candidate Key selected** by the designer as the official unique row identifier.
- **Cannot contain NULL** values; must be strictly unique.
- Only **one Primary Key** per table.
- Represented by an **Underlined Oval** in ER diagrams.
- **Example:** `Roll_No` is chosen as Primary Key from candidates `Roll_No` and `Email`.

4. Foreign Key

- An attribute in one table that **refers to the Primary Key** of another table.
- Enforces **Referential Integrity** — ensures related records exist in the referenced table.
- **Example:** `Roll_No` in `ENROLLMENT` table is a Foreign Key referencing `Roll_No` (PK) in `STUDENT` table.



★ Must-Write Points (for 10 marks)

1. Attribute types: Simple, Composite, Single-Valued, Multi-Valued, Stored, Derived.
2. Multi-Valued attribute uses Double Oval; Derived attribute uses Dashed Oval in ER diagrams.
3. Super Key = any set that uniquely identifies a row (may have redundant attributes).
4. Candidate Key = minimal Super Key (no redundant attributes); a table can have multiple.
5. Primary Key = the one Candidate Key selected; cannot be NULL; only one per table.
6. Foreign Key = references Primary Key of another table; enforces Referential Integrity.
7. Super Key $\supseteq$ Candidate Key $\supseteq$ Primary Key (each is a subset/specific case of the previous).

⚡ Quick Recall

Simple/Composite $\rightarrow$ Single/Multi-Valued (Double Oval) $\rightarrow$ Stored/Derived (Dashed Oval) $\rightarrow$ Super Key $\rightarrow$ Candidate Key (minimal) $\rightarrow$ Primary Key (chosen, no NULL) $\rightarrow$ Foreign Key (referential integrity)